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An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning

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ABSTRACT

The relentless evolution of phishing methodologies causes significant financial losses, data breaches, and reputational damages for individuals and organizations. Traditional detection methods struggle with limitations such as the inability to adapt to rapidly changing attack patterns and the computational inefficiency of frequent model updates. To address these limitations, we propose a novel framework with the following objectives: develop an adaptive framework integrating dynamic feature selection with online learning techniques, implement efficient drift detection mechanisms for real-time feature distribution monitoring, and validate performance improvements against conventional methods. Our proposed method employs Kolmogorov-Smirnov tests for drift detection alongside online Stochastic Gradient Descent for continuous model optimization. Validation using real-time phishing URLs from PhishTank demonstrates 99.85% accuracy with a minimal 0.067% false positive rate, substantially outperforming traditional approaches.

1 | Introduction

Phishing has emerged as one of the most prevalent and dangerous cyber threats today, responsible for billions of dollars in losses annually [1]. Cybercriminals employ a wide range of tactics, leveraging email, social media, SMS, and instant messaging platforms to disseminate fraudulent URLs that masquerade as legitimate websites [2–4]. By clicking on these malicious links, unsuspecting users may inadvertently disclose sensitive login credentials or incur substantial financial losses. Phishing attacks are indiscriminate in their targets, ensnaring everyone from everyday internet users to government officials. The increasingly sophisticated methods employed by attackers, who utilize advanced technologies to craft highly convincing fraudulent messages designed to deceive even the most vigilant and experienced individuals, make phishing particularly insidious [5]. Beyond financial motives, phishers often use phishing as a vector to distribute malicious software, including ransomware, further amplifying the threat by causing widespread disruption and data compromise [6].

As phishing techniques continue to evolve, so do strategies for defence against them. Organizations and individuals increasingly recognize the need for robust cybersecurity awareness training to educate users about the signs of phishing attempts and promote safe online behavior [7]. Advanced machine-learning

Abbreviations: KDE, Kernel Density Estimation; KS-test, Kolmogorov–Smirnov test; SGD, Stochastic Gradient Descent; URI, Uniform Resource Identifier; URL, Uniform Resource Locator.

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algorithms and AI-driven solutions are being implemented to analyze and identify suspicious URLS (Uniform Resource Locators) and emails in real time, providing an additional layer of protection [8]. Moreover, the development of multi-factor authentication (MFA) is gaining traction as a critical security measure [6], significantly reducing the likelihood of unauthorized access even if login credentials are compromised. However, despite these advancements, the arms race between cybercriminals and defenders persists, underscoring the necessity for continuous vigilance and adaptation in the fight against phishing [9, 10]. As technology advances, so must our detection and response methods, ensuring that users remain protected in an increasingly perilous digital landscape [11].

As cybercriminals continuously refine their tactics, traditional phishing detection methods face significant challenges in keeping pace with these rapidly evolving threats [8]. Machine learning-based approaches have emerged as a promising solution for the real-time detection of phishing URLs, owing to their efficiency in terms of training speed, reduced data requirements, and computational cost. Ensemble learning has gained traction for enhancing model performance across diverse applications, including recent work by Sharma et al. [12], which demonstrated the effectiveness of combining voting and stacking strategies in a multi-phase architecture. However, these conventional methods are often constrained by static, predefined feature sets and batch-learning frameworks, which impede their ability to adapt to emerging phishing patterns [13]. Batch learning methods, characterized by training models on large, fixed data sets at once, cannot accommodate real-time data, resulting in slow adaptation to new and evolving threats. This rigidity, with reliance on historical data and fixed thresholds, limits the model's capacity to detect novel phishing techniques, leading to higher false positive rates and undetected phishing attempts [14]. To address these limitations, there is a critical need for adaptive detection systems that can continuously learn from real-time data streams, ensuring more accurate and responsive detection in the face of sophisticated and dynamic phishing attacks.

Traditional models also encounter significant challenges due to the vast volume and the poor quality of features, the feature selection techniques employed, and dependence on third-party data sources. The excessive number of features can introduce inefficiencies, diminishing the model's speed and accuracy. Furthermore, reliance on external data sources hampers real-time applicability, limiting the system's ability to respond to emerging threats. These limitations underscore the pressing need for more adaptive, dynamic strategies to combat phishing in real time. The objective of this study is to develop an adaptable phishing attack detection system that overcomes the limitations of traditional approaches. We aim to build a system capable of real-time detection by harnessing online machine-learning techniques. This involves dynamically updating the detection model with new data to enhance its accuracy and responsiveness. By integrating advanced feature selection methods and continuous model updates, our system aims to stay ahead of evolving phishing tactics.

Our research emphasizes using URL-based features for phishing detection, offering a streamlined approach that identifies malicious URLs with minimal computational overhead. This method presents distinct advantages over traditional approaches reliant on website content, HTML features, visual similarity, or hybrid analysis. By focusing exclusively on URL features, we enable rapid extraction without the need to access or evaluate webpage content, significantly enhancing detection speed and responsiveness, which are critical for real-time applications. Furthermore, this approach is notably resource-efficient, avoiding the computational demands and extensive data sets typically required for content analysis or machine-learning models. Additionally, by circumventing the need to examine webpage content, we effectively address privacy concerns, as no personal or sensitive data is analyzed. These advantages position URL feature analysis as a highly efficient, privacy-preserving, and scalable solution for developing robust phishing detection systems.

Building on this foundation, this paper introduces a novel framework that integrates adaptive feature selection with online learning, enabling continuous learning from new data without requiring complete model retraining. The main novelty of this work lies in the integration of adaptive feature selection with online learning for real-time phishing URL detection. Specifically, we propose a two-stage feature selection pipeline. In the first stage, static feature selection is performed using Lasso regularization to filter out redundant and low-contribution lexical features, followed by Mutual Information (MI) to select the most informative attributes for classification. In the second stage, the Kolmogorov–Smirnov (K–S) drift detection method is applied to monitor distributional changes over time and to dynamically re-evaluate the optimal feature subset. This ensures both model efficiency and interpretability. To support real-time adaptation, we incorporate Online Stochastic Gradient Descent (SGD), enabling the model to continuously update its decision boundaries in response to evolving phishing tactics without the need for full retraining. Unlike many existing methods, our framework exclusively relies on lexical features extracted directly from URLs, eliminating the need for third-party data sources and making it lightweight and scalable. This novel combination of adaptive, drift-aware feature monitoring with online learning provides a robust and efficient defense against dynamic and sophisticated phishing threats.

The significant contributions of our study are:

- We developed an automated feature extraction process that parses URL strings to generate informative feature vectors, ensuring a scalable and efficient pipeline.
- We analyzed 18 critical URL features, revealing significant temporal distribution shifts in phishing characteristics and providing empirical evidence of feature drift to guide adaptive learning strategies.
- We developed an event-triggered feature re-evaluation mechanism and dynamic feature importance weighting with a novel integration of Kolmogorov-Smirnov drift detection and integrated Lasso regularization and Mutual Information (MI) for adaptive feature selection that dynamically refines the most relevant features.
- We developed an online learning approach with targeted mini-batch updates based on statistically significant distribution shifts, ensuring continuous model relevance, rapid

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adaptation to new phishing patterns, and high detection accuracy without full-scale retraining.

- We comprehensively evaluated our proposed model against state-of-the-art phishing detection methods. Our model consistently outperformed these existing approaches regarding accuracy and adaptability, demonstrating superior performance in real-time and evolving phishing scenarios.

The remainder of the paper is structured as follows: The related work is covered in Section 2, and the methodology is explained in Section 3. The experimentation is described in Section 4. Details about the results are provided in Section 5. The paper is concluded, and future works are discussed in Section 6.

2 | Relatedwork

This section explores notable research contributions and recently introduced machine learning-based methodologies for phishing attack detection. Phishing detection primarily employs supervised classification approaches, utilizing labeled data sets for model training. An effective solution in this domain necessitates high accuracy, minimal false positives, and reduced computational complexity to ensure real-time detection capability. The findings from the literature are synthesized and critically evaluated below. The summary of the literature findings is given in Table 1.

2.1 | Machine Learning-Based Phishing Detection

Ozgur Koray Sahingoz et al. [15] developed a machine learning-based phishing detection system, emphasizing Word Vectors, NLP-based features, and hybrid feature sets for real-time detection of phishing web pages. The data set, constructed from publicly available URLs, underwent preprocessing with three main components: a word decomposer module for analyzing compound words, a random word detection module to identify non-meaningful words, and a maliciousness analysis module to detect fraudulent intent in URLs. The authors evaluated several machine-learning algorithms using NLP-based, word-based, and hybrid features, including Naive Bayes, Random Forest, kNN, Adaboost, K-star, SMO, and Decision Tree. Among these, the Random Forest algorithm achieved the highest accuracy of 97.98% with the NLP-based features. However, the proposed method cannot detect tiny URLs and homoglyphs present in the URLs.

Andrei Butnaru et al. [16] employed supervised machine learning to detect phishing attacks using unique URL-based features. The paper introduces features focused on suggestive vocabulary, similarity, and Levenshtein distance. The proposed method is evaluated using an imbalanced data set with a realistic ratio of benign to phishing URLs. The approach excels with the Random Forest algorithm, achieving an F1 score of 98.86. The authors evaluated the model against Google Safe Browsing (GSB) on active phishing data sets, and it outperformed GSB. However, while the model is tested on an imbalanced dataset, its performance on diverse datasets remains unexplored. Additionally, the paper does not provide a detailed comparative analysis with other state-of-the-art phishing detection methods.

Jain and Gupta [17] proposed a machine learning-based client-side solution for phishing detection that leverages features extracted from URLs, hyperlinks, and identity keywords. Their approach identifies 19 distinct features from the URL and the webpage's source code. Experiments were conducted on a dataset comprising 2141 legitimate and 1918 phishing web pages, demonstrating a true positive rate of 99.39% and an overall detection accuracy of 99.09%. However, the experiments were conducted on a limited data set, and the model requires regular updates to adapt to new phishing techniques.

Shayan Abad et al. [18] propose a framework for evaluating machine-learning models for classifying malicious URLs. The authors extracted 16 URL features from the URLs in the Kaggle data set and employed instance selection methods and an ensemble of reduction techniques to identify the optimum feature set. The authors highlighted the role of the "has http" feature across all methods. Out of the implemented classifiers (DT, RF, SVM, and KNN), RF demonstrate superior performance, while SVMs offer competitive results despite longer training times. The study does not provide a detailed comparative analysis with other state-of-the-art phishing detection methods.

Jain and Gupta [19] presented a client-side approach for detecting phishing attacks by analyzing hyperlinks within the HTML source code of websites. The proposed method introduced unique hyperlink-specific features, categorized into 12 groups, used to train machine-learning algorithms. Experiments were conducted on a data set of 1428 phishing and 1116 benign pages. The logistic regression classifier achieved an accuracy of over 98.4%. However, the data set was relatively small and may not cover the full range of phishing attack variations. Additionally, since the method relies solely on the HTML source code, any modifications or obfuscations by attackers could reduce the model's accuracy.

Sumitra Das Gupta [20] proposed a hybrid feature-based machine-learning model designed for real-time phishing attack detection. This model integrates 15 URLs and 9 hyperlink features to provide a client-side solution. The study utilized a data set of 6000 URLs for experimentation, evenly divided between phishing and legitimate sites. Five classifiersÃćâĆňâĂİLogistic Regression, Decision Tree, SVM, XGBoost, and Random ForestÃćâĆňâĂİwere tested, with XGBoost achieving the highest accuracy of approximately 99.0%. However, the data set's limited scope may prevent the study from addressing the full range of phishing techniques and emerging tactics.

B.B. Gupta et al. [17] proposed a real-time phishing detection method based on nine lexical-based URL attributes. The nine features were ranked according to importance using different feature selection algorithms. The authors found that the proposed features are optimal and offer the best accuracy. They used various machine-learning classifiers to evaluate the suggested features, and the random forest method produced the highest accuracy (99.57%). While this approach demonstrated effectiveness, it does not address the adaptability of the model to evolving phishing tactics.

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TABLE 1 | Summary of key findings from existing literature on phishing URL detection.

Study	Feature set	Algorithms	Dataset	Accuracy	Limitations
Sahingoz et al. (2019) [15]	NLP-based features, word-based features, hybrid feature set	NB, RF, kNN, Adaboost, K-star, SMO, DT	Publicly available URL data set	97.98% (RF with NLP features)	Depends on static feature set, Cannot detect tiny URLs or homoglyphs.
Butnaru et al. (2020) [16]	Suggestive vocabulary, similarity, Levenshtein distance	Random Forest	Imbalanced data set with benign and phishing URLs	98.86% (F1 score with Random Forest)	Performance on diverse data sets unexplored; limited comparative analysis.
Jain et al. (2017) [17]	19 features from URL and webpage source code	Random Forest	2141 legitimate and 1918 phishing web pages	99.09%	Small data set; requires regular updates for new phishing techniques.
Abad et al. (2020) [18]	16 URL features	DT, RF, SVM, kNN	Kaggle URL data set	Best with RF; SVM competitive despite longer training	No detailed comparative analysis with state-of-the-art methods.
Jain and Gupta (2022) [19]	12 hyperlink-specific features	Logistic Regression	1428 phishing and 1116 benign pages	98.4%	Small data set; reliance on HTML can reduce latency.
Sumitra Das et al. (2022) [20]	15 URL features, 9 hyperlink features	LR, DT, SVM, XGBoost, RF	6000 URLs (half phishing, half legitimate)	99.0% (XGBoost)	Limited dataset may not cover all phishing techniques.
Gupta et al. (2020) [17]	9 lexical-based URL attributes	Random Forest	ISCX 2016 URL Data set	99.57% (RF)	Limited to nine fixed features, lacking adaptability.
Aljofey et al. (2021) [1]	URL character sequences, hyperlink, TF-IDF, HTML features	XGBoost	27,280 phishing and 32,972 benign URLs	96.76%	Language dependency; Dependence on features derived from HTML or page content hinder real-time performance.
Prasad et al. (2020) [21]	URL, HTML, visual similarity index	PhiUSIIL URL Dataset	BernoulliNB, PA, SGD	99.79% (pre-trained); 99.24% (incremental)	High resource demands: Retrieving HTML pages slows response time, posing challenges for real-time applicability.

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Ali Aljofey et al. [1] proposed a machine-learning method that leverages a hybrid feature set combining URL character sequences, hyperlink features, TF-IDF, and HTML features. The authors created a data set from publicly available URLs, including 27 280 phishing URLs with HTML codes and 32 972 benign pages. The findings suggest that the integrated feature set enhances classifier performance, achieving an accuracy of 96.76% with a 1.39% false positive rate on the data set. However, using plain text content-based features introduces language dependency, and HTML content-based features can degrade performance, making the model less suitable for real-time detection.

Machine learning-based applications have also been successfully deployed in other cybersecurity domains, such as deepfake detection. For example, Sharma et al. [12] proposed a VOTSTACK model that combines voting and stacking ensemble techniques with Decision Tree, Logistic Regression, and SVM classifiers, utilizing PCA and RFECV for optimized feature selection. Although not directly applied to phishing detection, its robust ensemble design and adaptive learning pipeline demonstrate the broader applicability of machine learning in addressing evolving digital threats, offering methodological insights that can inform phishing detection frameworks. Similarly, the concept of Digital Twin (DT) technology has emerged as a transformative approach in various industrial domains [4]. While primarily focused on industrial applications, the foundational principles of Digital Twin can also inform cybersecurity frameworks by enabling dynamic system modeling and threat response.

Recent advances in deep learning have also been applied to secure digital financial transactions. For instance, Khan et al. [11] proposed the FinSafeNet that combines Bi-Directional Long Short-Term Memory (Bi-LSTM), Convolutional Neural Networks (CNN), and a dual attention mechanism to analyze transaction data and detect security threats effectively. The model integrates an Improved Snow-Lion Optimization Algorithm (I-SLOA) for feature selection alongside Multi-Kernel Principal Component Analysis (MKPCA) with Nyström Approximation for efficient dimensionality reduction. Additionally, it employs advanced correlation measures and Joint Mutual Information Maximization to identify the most relevant features. Evaluated on the Paysim data set, FinSafeNet achieves a high accuracy of 97.8%, demonstrating its effectiveness in real-time transaction security.

Beyond traditional security domains, Qiao et al. [22] conducted a comprehensive review of biologically inspired covert communication systems, particularly focusing on underwater acoustic channels. While outside the phishing domain, this review underscores how nature-inspired mechanisms, and adaptive behaviors can inform the design of stealthy and resilient communication systems. These ideas could be extrapolated to phishing defense strategies, where adversaries mimic legitimate behavior covertly, necessitating equally adaptive and biologically inspired detection mechanisms.

Recent research has also explored secure and covert communication in other domains, which offer valuable insights into designing resilient communication frameworks, although not directly focused on phishing detection. For instance, Khan et al. [23] provides a comprehensive review of low probability detection-constrained underwater acoustic communication, highlighting how advanced signal processing and communication strategies can enhance stealth and reliability in adversarial environments. Such concepts could inspire future security mechanisms in phishing detection systems, especially evasion-resistant communication models.

2.2 | Adaptive-Online Learning-Based Phishing Detection

Conventional machine-learning methods typically utilize a batch learning approach. Batch learning, or offline learning, involves training a machine-learning model using the entire data set at once, leading to inefficiencies with large data sets due to high computational costs and long training times. This approach also lacks flexibility, as the model cannot easily adapt to new data or changes in data distribution without undergoing complete retraining. Additionally, managing and processing large volumes of data can be challenging, and the static nature of batch learning makes it unsuitable for real-time applications where timely updates are essential. Online learning overcomes the drawbacks of traditional batch learning, in that the model can be updated instantly and efficiently by an online model when new training data arrives. Only limited works are available in the field of online models for phishing detection. Our study explores the available methods in this niche, focusing on their effectiveness and limitations.

Prasad A et al. introduced PhiUSIIL [21], a phishing URL detection framework that employs a Similarity Index and Incremental Learning to detect visually similar phishing attacks. The framework continuously updates its knowledge base with new data through incremental learning, utilizing URL features, HTML features, and additional characteristics. Experimental results demonstrated PhiUSIIL's detection accuracy, achieving 99.24% with a fully incremental training approach and 99.79% with a pre-training approach. However, the proposed methods encounter significant challenges in terms of latency, resource intensity, making them less suitable for real-time phishing detection.

2.3 | Research Gap

Despite significant advancements in phishing detection research, several limitations highlight the need for more adaptable models. While many studies have focused on identifying effective classifiers, a notable lack of research is dedicated to determining the most effective feature sets. Lightweight and computationally efficient feature sets are crucial for enhancing the practicality of phishing detection systems in real-time applications. Many studies rely on static feature sets that struggle to capture evolving phishing tactics, leading to higher false positives and missed detections. Resource dependency is another issue, as some models require substantial computational power or large data sets, limiting their applicability in resource-constrained environments like IoT devices. Additionally, many models lack real-time adaptability, often requiring periodic retraining, which delays the response to new threats. The continuously evolving threat landscape necessitates models that can quickly adjust to new phishing tactics. Therefore, adaptable models that dynamically update

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feature sets adjust to real-time data streams, optimize resource usage, and improve detection accuracy are crucial to robust protection against phishing threats.

3 | Methodology

This section provides a detailed overview of the proposed system for detecting phishing URLs. To ensure real-time adaptability in phishing URL detection, the proposed framework includes a dynamic feature selection and online learning framework that continuously refines itself as new data arrives, as depicted in Figure 1. This approach integrates LASSO-based feature selection, mutual information ranking, and online learning via Stochastic Gradient Descent (SGD), enabling the system to remain resilient against evolving phishing strategies.

3.1 | URL Collection

This study utilizes URLs from three prominent data sets to ensure a diverse and comprehensive collection of phishing and legitimate URLs, as shown in Table 2. The primary sources include the Kaggle data set [24], the ISCXURL data set [25], and the PhiUSIIL [21] URL Dataset. From the Kaggle data set, 10 000 phishing URLs and 10 000 benign URLs were selected. The ISCXURL collection comprises approximately 35 300 benign URLs obtained from top Alexa websites and 9964 phishing URLs sourced from OpenPhish, a real-time database of active phishing sites. To create a balanced data set for initial model training, we selected 10 000 benign URLs and 9964 phishing URLs from the ISCX collection, totaling 39 964 URLs. Additionally, we incorporated the PhiUSIIL Phishing URL Data set, a substantial and up-to-date resource containing 134 850 legitimate and 100 945 phishing URLs. This data set includes a significant portion of recently observed URLs, making it particularly valuable for evaluating the performance of phishing detection models against contemporary threats.

3.2 | Preprocessing

Raw URLs are often noisy and unstructured, containing missing values, redundant components, or irregular formatting that can

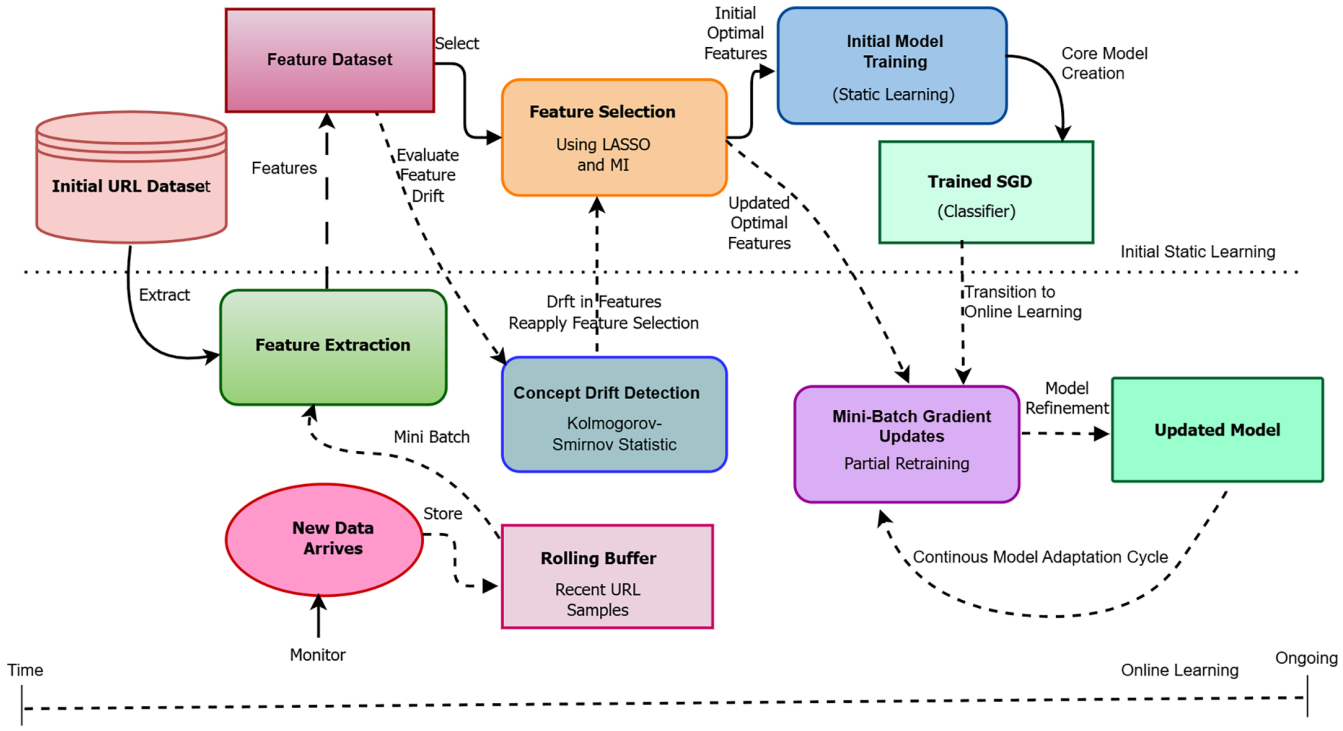


FIGURE 1 | Proposed method framework.

TABLE 2 | Details of the data sets used for phishing URL detection.

Dataset	Source	Phishing URLs	Benign URLs	Total URLs
Kaggle [24]	Kaggle Archive	10 000	10 000	20 000
ISCXURL [25]	OpenPhish	9,964	—	9,964
	Alexa Top Websites	—	10 000	10 000
PhiUSIIL [21]	Phishtank, Openphish, Malwareworld	100 945	—	100 945
	Open PAgeRank	—	134,850	134,850
Total (Used in Study)		**120 909**	**154 850**	**275759**

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hinder accurate feature extraction. We applied a comprehensive preprocessing pipeline to enhance the quality and consistency of the data before feature extraction. First, URLs are standardized by converting them to lowercase. Next, tokenization decomposes each URL into its fundamental parts to facilitate detailed analysis. A typical URL consists of several components, such as the scheme (e.g., http, https), domain name (e.g., example.com), subdomains (e.g., mail.example.com), path (e.g., /login/index.html), and query parameters (e.g., ?id = 123). Tokenization involves separating these components and further breaking them into smaller units, such as individual words, numbers, delimiters (like slashes or hyphens), or special characters. This step is essential because malicious URLs often embed suspicious patterns or keywords within these structures that can indicate phishing behavior. Following tokenization, feature encoding is applied to transform categorical components-such as scheme, file extensions, to numerical values using one-hot encoding. These steps ensure that the extracted features are consistent, relevant, and informative.

3.3 | Feature Extraction

Feature extraction is a critical component of our phishing detection framework, enabling real-time analysis of raw URLs without reliance on external databases or APIs. To build a robust detection system, we extracted a diverse set of lexical, structural, keyword-based, and obfuscation-related features. Our feature selection was guided by an extensive review of existing research on phishing detection and cybersecurity studies, ensuring the inclusion of well-established and novel indicators of phishing attempts.

Traditional phishing detection systems primarily focus on features like URL length, token count, and suspicious keywords. While these surface-level characteristics provided adequate detection in earlier phishing campaigns, they suffer from significant limitations: they are easily manipulated by attackers, fail to capture sophisticated obfuscation techniques, and generate high false positive rates when legitimate URLs adopt complex structures. Moreover, these features become increasingly ineffective as attackers deliberately craft URLs to match statistical profiles of benign websites. To address these shortcomings, we introduced additional features targeting evolving phishing tactics and sophisticated obfuscation methods, including homoglyph detection, URL encoding analysis, redirect chain evaluation, and JavaScript execution tracking. Our feature engineering approach was guided by an analysis of emerging evasion techniques observed in contemporary phishing campaigns.

The inclusion of redirect chain complexity was particularly significant, as our preliminary analysis revealed that sophisticated phishing attempts employed multiple redirection layers to evade detection. By quantifying both the depth of redirects and the entropy between domain transitions, this feature provides critical insights into potential malicious infrastructure designed to conceal the attack's origin. Similarly, homoglyph detection addresses the increasing prevalence of Unicode-based deception, while URL encoding analysis captures obfuscation patterns that conventional lexical analysis might miss. JavaScript execution tracking was incorporated after identifying that dynamic content loading has become a preferred method for delivering payloads post-detection. These enhancements provide a multi-layered approach, allowing the detection of phishing URLs through multiple perspectives and mitigating the limitations of conventional detection techniques. The features utilized in this study are comprehensively outlined in Table 3.

We extracted the redirect chain, which represents the sequence of redirections a URL undergoes before reaching its final destination. Phishers manipulate redirect chains to obscure malicious destinations, using multiple redirects to bypass security filters and deceive users into believing they are accessing a legitimate site. The redirect chain analysis feature identifies complex patterns without executing potentially malicious code. We introduced a redirect chain scoring mechanism that evaluates the complexity of URL redirections to detect potential phishing attempts. This mechanism assigns a cumulative risk score based on key redirection attributes, including the number of hops, domain mismatches, TLD changes, HTTPS downgrades, and looping behavior. Higher scores indicate a greater likelihood of phishing activity. Unlike traditional detection methods, this approach effectively identifies cloaking techniques and deceptive redirect patterns without executing potentially malicious scripts, making it efficient for real-time phishing detection.

We employed an extensive keyword-matching approach, surpassing conventional phishing detection systems that typically rely on only three to four keywords. Phishers strategically embed deceptive keywords such as "login," "verify," "account," "secure," and "billing" within URLs to mimic legitimate services and lure users into interacting with fraudulent sites. In addition to standard terms, we expanded our keyword set to include a broader range of phishing-related phrases such as "secure," "payment," "billing," "free," "prize," "password," "support," "authenticate," "service," "reset," "unlock," "transaction," "claim," "winning," "alert," "suspend," and "reactivate". This enriched vocabulary enhances the model's ability to detect a wider array of phishing attempts, particularly those leveraging social engineering tactics through financial transactions, security alerts, or deceptive offers.

We also checked the presence of homoglyphs in URLs, leveraging a comprehensive set of homoglyph characters to identify their presence in URLs. Homoglyphs are visually similar characters from different scripts (e.g., 'o' (Cyrillic) vs. 'o' (Latin)) that attackers use to create deceptive domain names resembling legitimate ones. Our method performs an extensive character similarity analysis, enabling the detection of these subtle substitutions commonly exploited in phishing attacks. By recognizing homoglyph-based obfuscation techniques, our approach enhances detection accuracy and mitigates the risk of visually deceptive phishing URLs.

Phishers incorporate JavaScript and Data URIs (Data Uniform Resource Identifiers) in URLs to execute malicious scripts directly in a web browser. These obfuscated JavaScripts can automatically redirect users, steal credentials, or perform harmful actions without user interaction. On the other hand, data URIs allow attackers to embed entire malicious payloads-such as HTML login forms or encoded phishing scripts-directly within a URL, bypassing traditional security checks. Since these attacks do not rely on external domains or traditional web requests, they are harder to detect using conventional blacklist-based methods. By

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TABLE 3 | URL features and their descriptions.

Feature category	Feature	Description
Basic URL characteristics	URL Length	Total character count in the URL.
	Domain Length	Character count in the domain portion.
	Path Length	Character count in the path portion.
	Token Count	Number of segments separated by delimiters.
	Number of Subdirectories	Deep directory structures may hide malicious content.
	URL Shortening	Use of shortened URLs to hide the true destination.
Suspicious elements	Special Character Count	Frequency of characters like @, −, _, =, %.
	Number Count	Frequency of numeric characters.
	Suspicious TLD	Presence of uncommon or suspicious TLDs.
	Has Port	Unusual ports may indicate suspicious activity.
	Entropy	Higher entropy suggests obfuscation techniques.
	Is IP	Direct IP addresses bypass domain-based blacklists.
	Multiple Redirects	Use of multiple redirections to obscure the final destination.
	Suspicious File Extensions	Unusual or executable file types can indicate phishing.
Advanced Detection	Homoglyph Detection	Identification of visually similar character substitutions.
	URL Encoding Analysis	Detection of excessive or suspicious encoding patterns.
	JavaScript/Data URI Detection	Identification of embedded scripts or encoded payloads.
	Non-alphanumeric Characters	Presence indicates obfuscation techniques.
	Entropy	Measures randomness in URLs to detect obfuscation.
Keyword Analysis	Suspicious Term Frequency	Presence of terms like login, verify, account, secure, payment, billing, free, prize, password, support, service, authenticate, reset, unlock, transaction, claim, winning, alert, suspend, and reactivate
	Brand Name Proximity	Detection of brand names in suspicious contexts.
Redirect Analysis	Redirect Chain Complexity	Analysis of URL redirection patterns.
	Domain Mismatch	Inconsistency between display and actual domains.

including JavaScript/Data URI detection as a feature, our model can identify suspicious patterns in URLs that indicate script execution, encoded payloads, or attempts to bypass security mechanisms, thereby improving the detection of modern, script-based phishing threats.

Unlike traditional studies that often utilize limited feature sets, our approach encompasses a broader range of lexical features, ensuring a more thorough and effective detection system. To facilitate this, we developed an automated feature extraction process that parses URL strings and extracts these key features. This automated approach generates informative feature vectors, which are then used as input for machine-learning algorithms. Our study identifies the most relevant lexical features independently of third-party sources. The feature extractor systematically retrieves these lexical features from pre-processed URLs and records them in a CSV file, creating a comprehensive data set of features for analysis and model training.

3.4 | Adaptive Feature Selection

The proposed phishing URL detection framework integrates adaptive feature selection with online learning to enhance detection capabilities in real-time environments. This adaptive feature selection is a critical component of our phishing URL detection framework, ensuring that the model remains resilient to evolving attack strategies while maintaining computational efficiency. Unlike traditional static feature selection, our approach dynamically refines feature importance based on newly arriving data, leveraging a combination of LASSO regularization, mutual information ranking, and online learning. The process consists of three key stages: initial feature selection, continuous monitoring, and dynamic adaptation.

Initial Feature Selection: In this phase, the LASSO (Least Absolute Shrinkage and Selection Operator) regularization is applied to select the most informative lexical features of phishing URLs. LASSO enforces sparsity in feature selection by assigning zero weights to less relevant attributes, effectively reducing dimension while retaining only predictive features. Mathematically, LASSO minimizes the following objective function:

$$\min_{w}\left(\sum_{i=1}^{n}(y_i - w^T x_i)^2 + \lambda\sum_{j=1}^{p}|w_j|\right) \quad (1)$$

where:

- y_i is the target label (phishing or benign).
- x_i is the feature vector.

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- w_j are the feature weights.
- λ is the regularization parameter, controlling the degree of sparsity.

As λ increases, more feature weights shrink to zero, automatically discarding irrelevant features.

After LASSO filters out irrelevant features, we apply the Mutual Information (MI) ranking to refine the selection further. Unlike LASSO, which is based on linear relationships, MI captures non-linear dependencies between features and the target variable. It measures how much knowing the value of a feature reduces uncertainty about the label (phishing or benign). Features with high MI scores provide strong discriminatory power and are ranked higher.

Mathematically, MI is given by:

$$MI(X,Y) = \sum_{x\in X}\sum_{y\in Y} P(x,y)\log\frac{P(x,y)}{P(x)P(y)} \tag{2}$$

where:

- $P(x,y)$ is the joint probability of feature X and the label Y.
- $P(x)$ and $P(y)$ are the marginal probabilities of X and Y, respectively.

Mutual Information (MI) is applied after LASSO because it captures non-linear relationships between features and phishing labels, which LASSO may overlook due to its linear nature. MI ranks features based on their predictive contribution, prioritizing those that offer the most information gain. Additionally, MI helps ensure that even if LASSO retains a feature, it is genuinely relevant for classification, further refining the feature selection process and enhancing the model's effectiveness.

Continuous Monitoring: To maintain high detection accuracy, the system implements a continuous monitoring and feature re-evaluation mechanism. This ensures the model remains adaptive and responsive to emerging phishing tactics by integrating real-time data-driven adjustments. The statistical properties of the features used for classification change over time due to the dynamic nature of phishing tactics. If certain lexical or structural patterns in phishing URLs change, the initially selected features may no longer be the most relevant. The system employs concept drift detection techniques to monitor changes in feature distributions. If a significant deviation is detected, it indicates that the importance of certain features has changed, necessitating re-evaluation and adaptation of the feature set.

Dynamic Adaptation: Since retraining a model from scratch with every new phishing trend is computationally expensive and impractical in real-time environments, an efficient way to incorporate new knowledge is required. When significant variations in feature importance are detected, the model undergoes a partial retraining process using Stochastic Gradient Descent (SGD). Rather than performing full retraining, this approach selectively updates feature weights while reapplying the feature selection mechanism. LASSO regularization is re-executed during this phase to remove outdated features and incorporate new ones dynamically, as given in Figure 2.

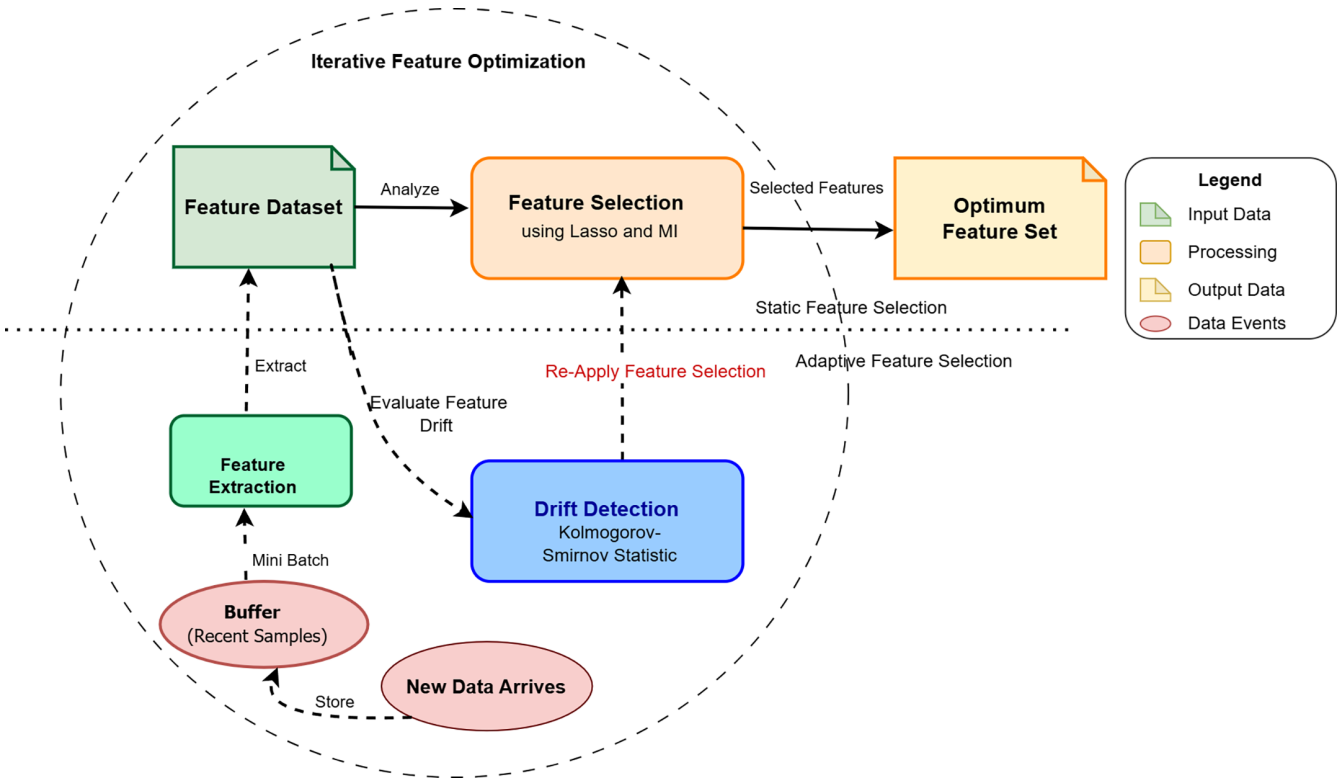


FIGURE 2 | Adaptive feature selection.

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ALGORITHM 1 | Adaptive Feature Selection Pipeline for Phishing URL Detection.

Input: SGD Model $\mathcal{M}$ Streaming URL dataset D with feature set $\mathcal{F}$, batch size B, threshold τ
Output: Adaptively selected feature set $\mathcal{F}^*$
Initialize: Feature importance history $\mathcal{H}$, feature drift monitor D_f
Step 1: Initial Feature Selection
Compute Lasso-derived feature weights: $\boldsymbol{w} = |\mathcal{M}.\text{coef}|$
Compute Mutual Information (MI) scores: $\boldsymbol{I} = \text{MI}(X_B, y_B)$
Compute combined feature importance: $\boldsymbol{S} = \boldsymbol{w} + \boldsymbol{I}$
Rank features: $\mathcal{F}_{ranked} = \text{Sort}(\mathcal{F}, \boldsymbol{S})$
Step 2: Concept Drift Detection and Continuous Monitoring
while New batch $(X_B, y_B) \in \mathcal{D}$ arrives **do**
    for each feature $f \in \mathcal{F}^*$ **do**
        Compute drift metric $\mathcal{D}_f(f)$ using Kolmogorov-Smirnov (KS) test
        if $\mathcal{D}_f(f) > \delta$ (drift detected) **then**
            Mark f as obsolete and remove from $\mathcal{F}^*$
        end if
        Set adaptive threshold: $\tau = \text{Percentile}(\boldsymbol{S}, 75)$
        Select top-ranked features: $\mathcal{F}^* = \{f \in \mathcal{F} \mid S(f) \geq \tau\}$
    end for
    Store latest feature scores: $\mathcal{H} \leftarrow \mathcal{H} \cup \boldsymbol{S}$
end while
return $\mathcal{F}^*$

The Adaptive Feature Selection Pipeline presented in Algorithm 1 is designed to dynamically select and update features in a real-time phishing URL detection system. The algorithm begins by initializing an online Lasso model, a feature importance history, and a drift monitor. As new URL batches arrive, it updates feature importance using Lasso-based weight coefficients and Mutual Information (MI) scores, creating a ranked feature list. An adaptive threshold is then applied to retain the most significant features. To ensure relevance, the algorithm detects feature drift using the Kolmogorov-Smirnov (KS) test and removes features that exhibit substantial distribution changes. Additionally, it incorporates a feature space expansion mechanism that introduces new features based on emerging phishing attack patterns. Finally, the selected features are stored in a historical log, ensuring continuous adaptation and refinement of the feature set for improved phishing detection accuracy. This approach enhances the model's resilience against evolving threats and maintains its effectiveness in a real-time cybersecurity environment.

3.5 | Online Machine Learning Model

To ensure real-time adaptation and responsiveness to evolving phishing threats, we employ an online machine-learning approach that continuously refines the model as new data arrives. This methodology consists of initial model training and incremental model updates, enabling efficient and scalable phishing detection in dynamic environments. The process begins with an offline training phase, where a baseline model is established using a data set consisting of optimal features obtained after feature selection, features (X), and target labels (y). The trained model undergoes an initial performance evaluation using a validation set, assessing key metrics such as accuracy, precision, recall, and F1-score.

The core of our approach lies in its online learning capability, where the model is incrementally updated as new phishing URLs are encountered, as depicted in Figure 3. Instead of undergoing periodic retraining, which is computationally expensive, the SGD classifier updates its weights dynamically using mini-batches of new data. We initialize a buffer for storing recent phishing and benign URL samples, facilitating incremental model updates while preventing overfitting. The buffer maintains a rolling window of the most recent observations, ensuring that the model learns from the latest data patterns while retaining historical knowledge. Concept drift detection is conducted using statistical tests, and the Kolmogorov-Smirnov test (KS-test) is used to detect a significant drift. Concept drift is a phenomenon where the statistical properties of phishing data change over time. The feature selection mechanism is re-executed to incorporate the most recent feature distributions. Each incremental update is followed by a rigorous performance evaluation. Our approach guarantees that the model remains current and responsive by simulating new data arrivals and incorporating them into the model through ongoing updates.

4 | Experimentation

All experiments were executed on a high-performance system with a Ryzen 7 processor, 16 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU. This setup ensured that the framework could process large URL data sets rapidly and perform real-time model updates efficiently. The proposed framework was implemented in Python, leveraging several key libraries to facilitate efficient data processing and model training. Scikit learn was utilized for implementing machine-learning algorithms and feature selection techniques, while NumPy and Pandas supported data manipulation tasks. Results were visualized using Matplotlib and Seaborn, and SciPy was employed for statistical testing. The complete source code and experimental setup are available at: https://github.com/sruthi-krishnan/Drift-Aware-Feature-Monitoring-and-Online-learning-.git.

We conducted a comprehensive analysis of URL feature distributions using 'Old' URLs-random samples drawn from the ISCX URL, Kaggle, and PhiUSIIL data sets and 'New' URLs obtained from the PhishTank database, collected between June 2024 and January 2025. Figure 4 presents kernel density estimation (KDE) plots comparing the distributions of 18 critical URL features between earlier ("Old") and more recent ("New") phishing samples. The x-axis represents the lexical feature, and the y-axis represents the density, which is a smoothed estimate of the probability distribution of that feature. These 18 features were specifically chosen to highlight significant shifts in URL characteristics over time. The selection was data-driven-based on features that exhibited notable distributional changes when comparing historical and recent data sets, as visualized through KDE plots. The visualizations reveal substantial temporal shifts in feature distributions, providing empirical evidence for the foundational premise of our proposed adaptive detection framework.

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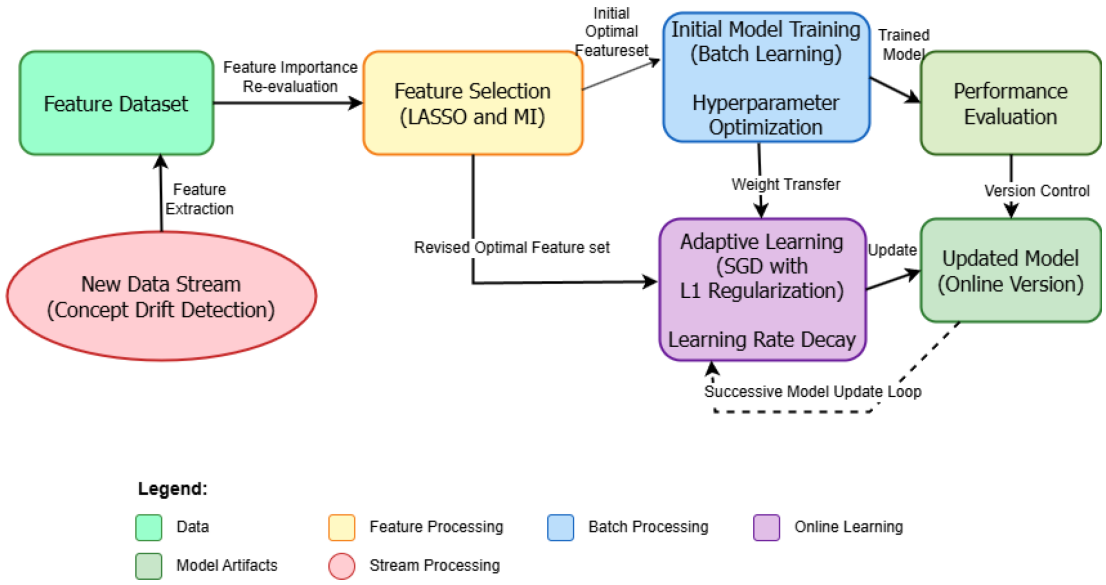


FIGURE 3 | Online learning pipeline.

Domain length distributions exhibit clear convergence patterns in newer samples, with both data sets showing similar multimodal characteristics. Entropy distributions demonstrate significant consolidation in recent samples, with higher density peaks indicating evolutionary changes in information content. The special character distributions (num_dots, num_slashes, num_special_chars) show marked shifts, with newer samples displaying concentrated distributions with reduced variance compared to older samples. Token-based features (e.g., longest_path_token_length) similarly reveal structural standardization in recent phishing attempts. Binary features (has_free, has_secure) show diminished presence in newer samples, suggesting tactical adaptation by attackers. The url_encoded and num_tlds distributions highlight further refinement in obfuscation techniques. These observed distribution shifts substantiate the need for our proposed adaptable feature selection mechanism and online machine- learning approach, as they empirically demonstrate that the discriminative power of individual features evolves significantly over time.

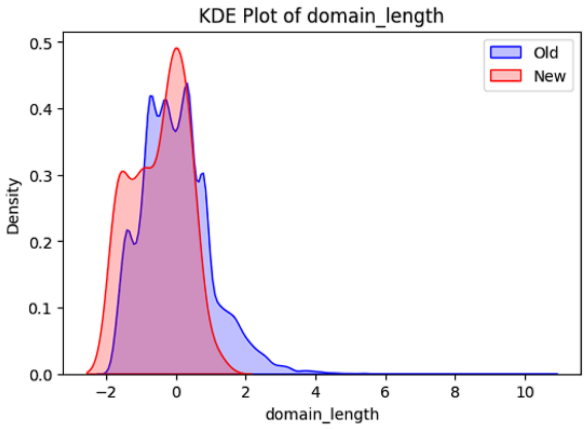
The overall experimental setup for our phishing URL detection framework consisted of two primary phases: an initial offline training phase and an online incremental learning mechanism. The process began with initial feature extraction, followed by LASSO regularization and Mutual Information ranking to identify the most relevant features. These selected features were used to train the initial model.

For the batch learning phase, we applied feature selection using LASSO regression, followed by Mutual Information (MI) analysis to optimize the feature set for URL classification. This two-stage approach allowed us to identify the most discriminative features while reducing dimensionality and mitigating overfitting risks. The LASSO regression effectively eliminated redundant features by imposing L1 regularization, while the subsequent MI analysis quantified the statistical dependence between remaining features and the target classification, further refining our selection. The optimum features retrieved by this feature selection process are visualized in the co-occurrence networks shown in Figure 5, illustrating how these features interact differently in malicious versus benign URLs. These networks validate our feature selection methodology and provide insights into the underlying patterns that distinguish legitimate web resources from potentially harmful ones. The network topology highlights feature relationships that might be overlooked when examining features in isolation, demonstrating the value of our approach in capturing complex patterns characteristic of URL-based threats.

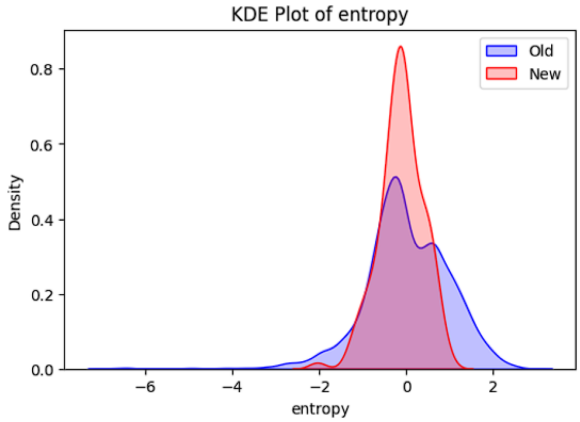
The feature co-occurrence network nodes represent individual URL features, with edge thickness indicating the strength of co-occurrence between features. Our analysis demonstrates that malicious URLs exhibit strong clustering among security-sensitive features. Specifically, we observe a dense subnetwork connecting suspicious_file_extensions, has_admin, has_login, and url_shortening features, suggesting these attributes frequently appear together in malicious URLs. This pattern aligns with known attack vectors where deceptive URLs attempt to appear legitimate while containing suspicious components. The network also reveals that structural features like url_length and num_delimiters_path share strong co-occurrence relationships, indicating that malicious URLs often employ complex path structures, likely to obfuscate their true nature or to exploit parsing vulnerabilities in browsers and security tools. Notably, the is_ip feature forms strong connections with scheme_http and suspicious_file_extensions, revealing that attackers frequently use direct IP addresses rather than domain names when deploying malicious content, particularly when using unencrypted HTTP connections.

The benign URL feature co-occurrence network displays markedly different patterns than its malicious counterpart. This network is characterized by more evenly distributed connections between feature nodes, reflecting the natural diversity of legitimate web resources. Our analysis shows benign URLs demonstrate strong co-occurrence between scheme_https and standard structural elements such as moderate url_length

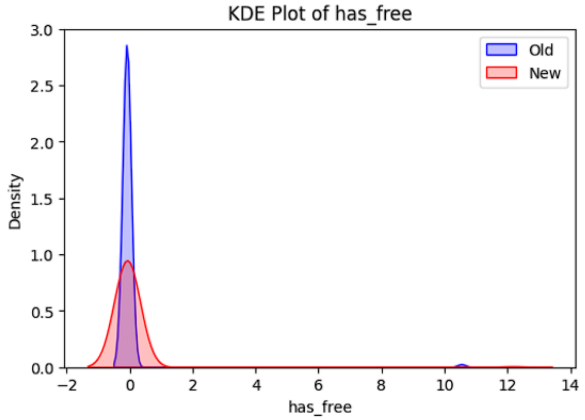
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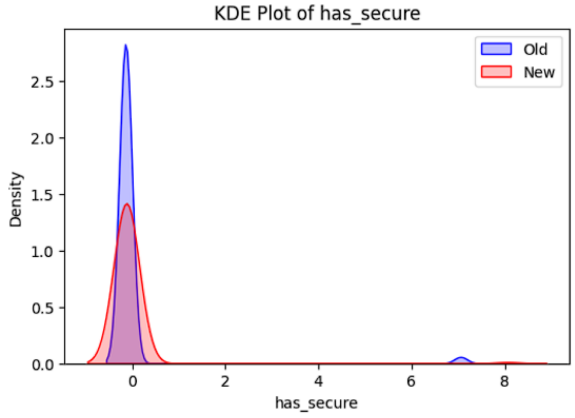
(a) Longer domains replacing traditional short phishing URLs



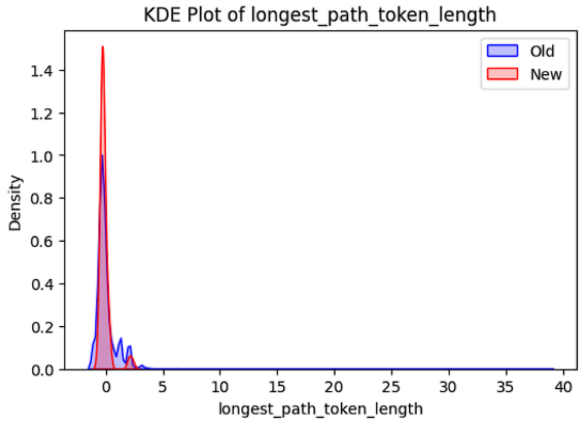
(b) Higher entropy values indicating more random character combinations



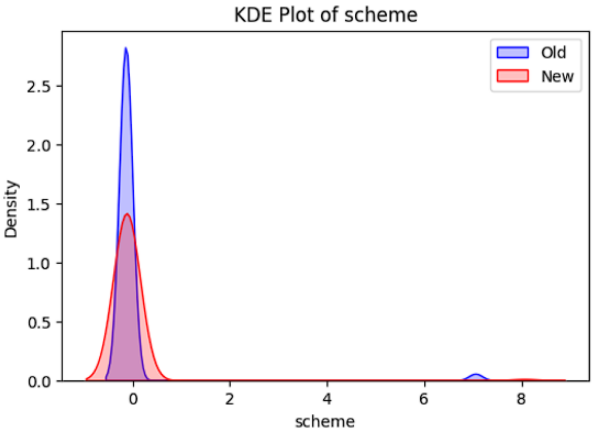
(c) Decreased usage of 'free' as bait in newer URLs



(d) Reduced 'secure' keyword impersonation tactics



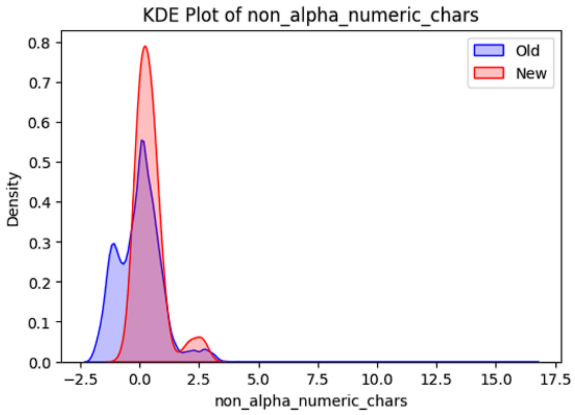
(e) Shorter Path Tokens in Recent Phishing URLs Mimicking Legitimate Site Structures



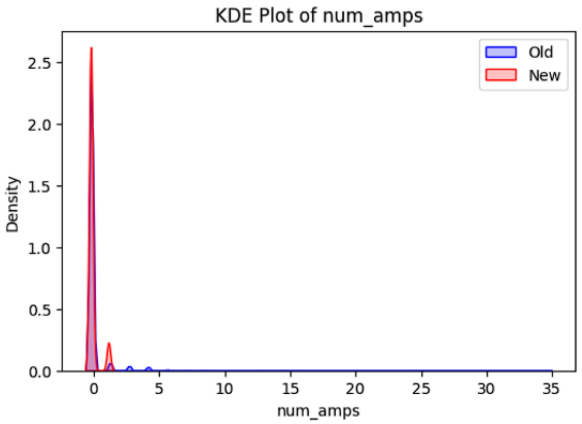
(f) Rising Adoption of HTTPS in Phishing URLs to Exploit User Trust Indicators

FIGURE 4 | Feature distribution: Old URLs (Random samples from ISCX, Kaggle and PhiUSIIL) vs. New URLs (From Phishtank Collected between June 2024 and Jan. 2025).

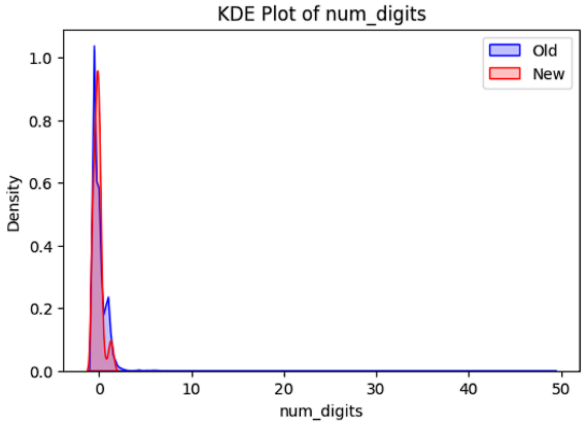
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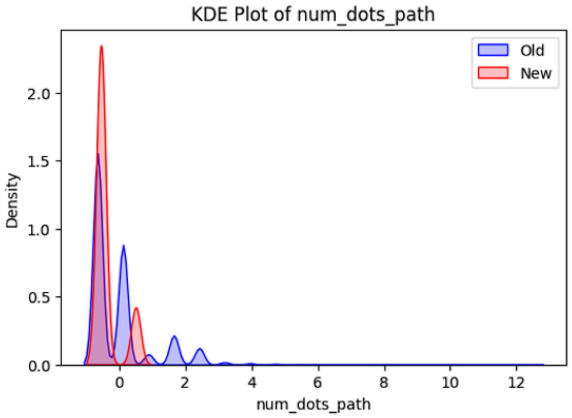
(g) Increased alphanumeric character obfuscation techniques



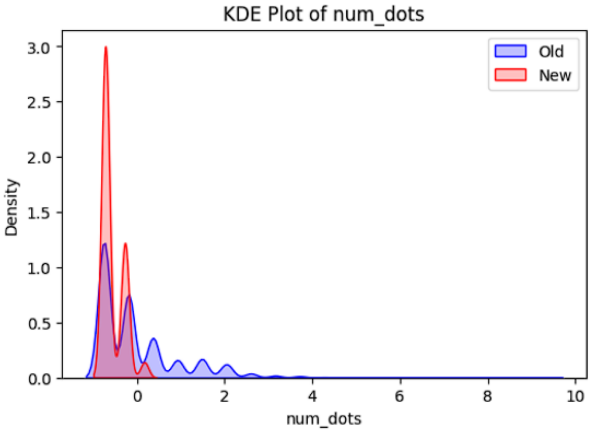
(h) Strategic manipulation of query parameters via ampersands (&)



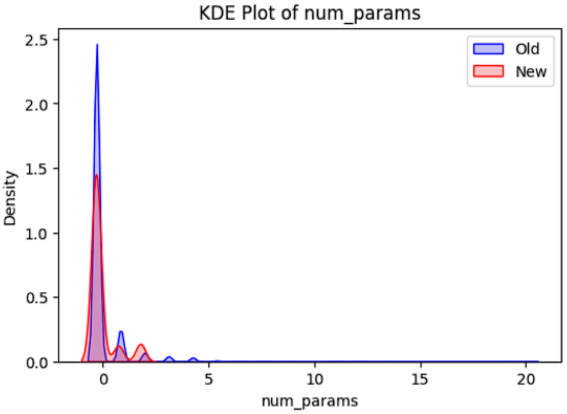
(i) Increasing digit frequency to create more random-appearing URLs



(j) Increased Use of Dots in URL Paths to Obfuscate Subdirectory Structures



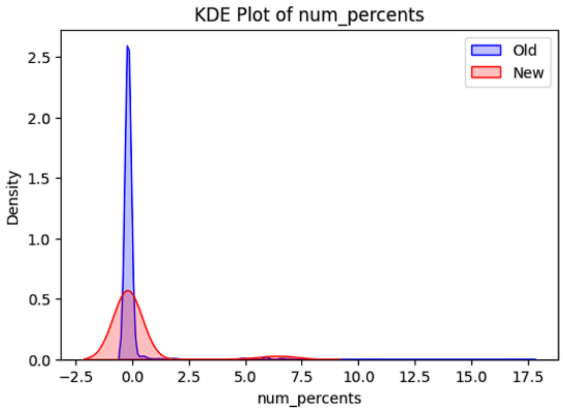
(k) Elevated Dot Counts in Domain Names of Newer URLs for Evasion



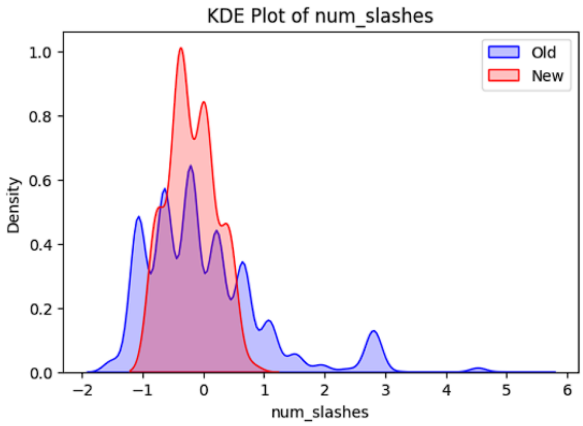
(l) Lower Query Parameter Density in Recent URLs to mimic legitimate link structures

FIGURE 4 | (Continued)

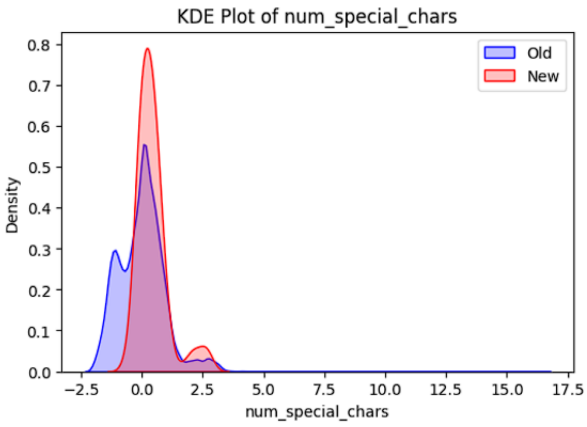
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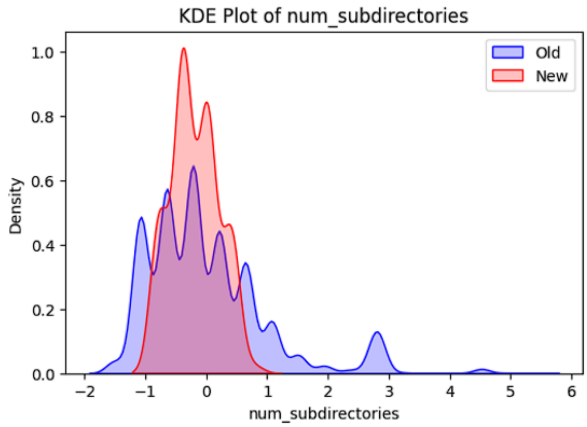
(m) Increased percent-encoding (%) for character obfuscation



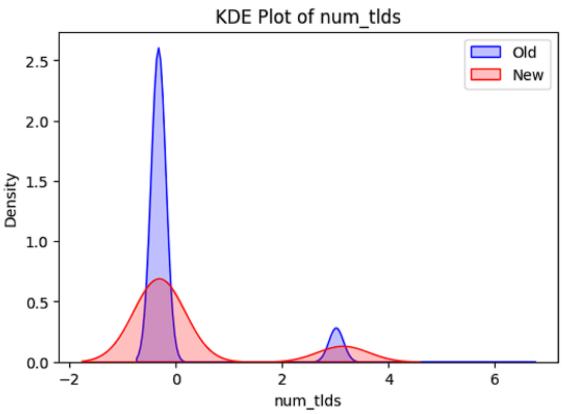
(n) Tactical use of slashes to create deeper, more complex URL structures



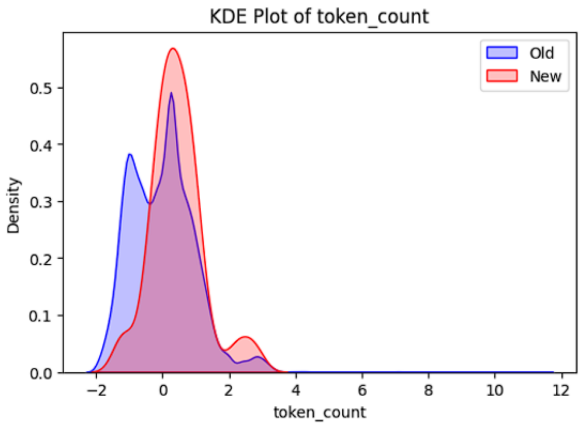
(o) Increased use of special characters for obfuscation



(p) Longer query strings (Subdirectories) enabling advanced payload delivery



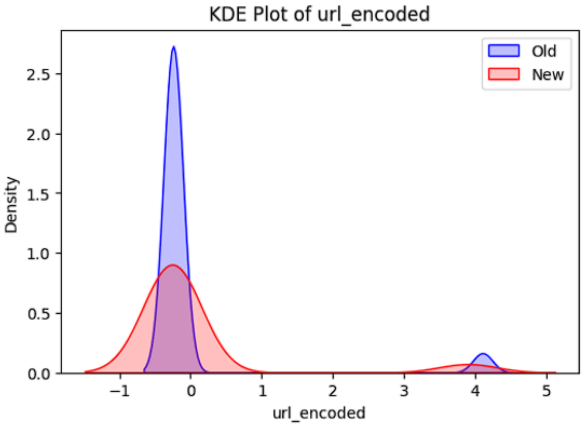
(q) Increased TLD variety for evading domain reputation systems



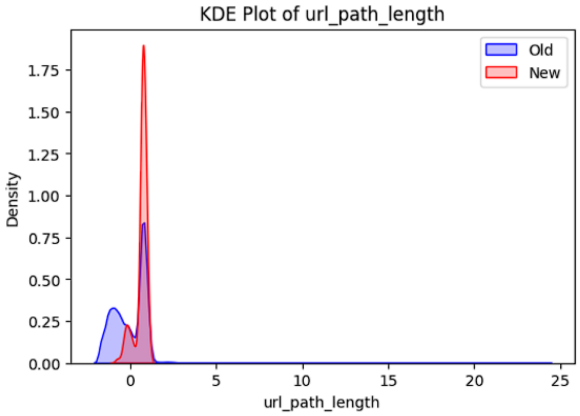
(r) More URL tokens reflecting complex path structures to mimic legitimate sites

FIGURE 4 | (Continued)

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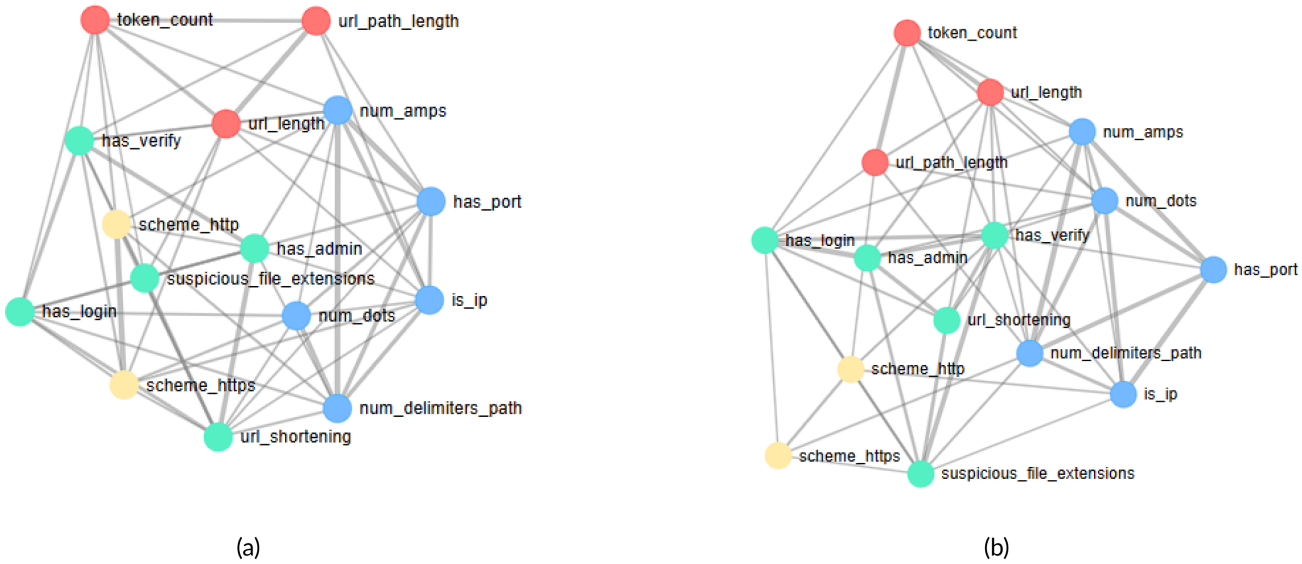


(s) Higher prevalence of URL encoding to disguise malicious parameters



(t) Shorter path lengths suggesting streamlined attack landing pages

FIGURE 4 | (Continued)



(a) (b)

FIGURE 5 | Feature co-occurrence network of optimal features. (a) Malicious URLs, (b) Benign URLs.

and url_path_length. This pattern indicates the widespread adoption of secure connections for legitimate websites, typically constructed with conventional and predictable URL structures. The network reveals weaker connections between security-sensitive features, with has_login and has_verify showing some co-occurrence but not clustering as tightly as in malicious URLs. When these features appear together in benign URLs, they are more likely to co-occur with scheme_https, reflecting legitimate security practices. The token_count and num_dots features show moderate co-occurrence in benign URLs, typically representing legitimate subdomains and conventional URL construction patterns rather than obfuscation attempts. The distinct topological differences between malicious and benign feature networks provide valuable insights for URL classification systems. The clustering patterns in malicious URLs suggest that detection systems could benefit from examining feature combinations rather than individual features in isolation. These network visualizations support our feature selection approach by highlighting which features provide complementary rather than redundant information.

In the offline phase, we trained the base model using Stochastic Gradient Descent (SGD). Since SGD inherently supports various linear models, we employed its implementations for Linear Regression, Support Vector Machines (SVM), and the Perceptron algorithm. Each model was initially trained using the optimal feature set identified through LASSO and MI, with training conducted on an 80–20 train-test split. To handle potential class imbalances in the data set, class weights were set to 'balanced', ensuring a fair representation of both phishing and benign URLs. An adaptive learning rate was employed to optimize convergence, allowing the model to dynamically adjust step sizes during training. Training was performed with a maximum of 1000 iterations, and early stopping was enabled using a 10% validation set, which halted training if no improvement was detected. This batch training phase established a robust initial model before

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TABLE 4 | Hyperparameters for offline and online learning phases.

Component	Parameter	Value
Feature selection	LASSO regularization strength (λ)	0.0001
	Maximum iterations	1000
Offline training (SGD)	Learning rate schedule	adaptive
	Initial learning rate (η_0)	0.01
	Maximum iterations	1000
	Validation fraction	0.1
	Early stopping	enabled
	Class weight	balanced
Online learning	Buffer size	1500
	Mini-batch size (B)	500
	Drift detection threshold (δ)	0.05
	Drift detection method	Kolmogorov-Smirnov test
Experimental setup	Random seed	42
	Train-test split ratio	80:20

transitioning to the online learning phase. The models were evaluated using standard performance metrics, including accuracy, precision, recall, and F1 score. After deployment, the models were updated online, and their performance was re-evaluated to assess adaptability in real-time environments. To facilitate reproducibility, we explicitly defined all stochastic and optimization parameters, as illustrated in Table 4.

To facilitate real-time updates, we collected streaming phishing URL data from PhishTank using its API, which provides continuously updated, verified phishing URLs. The retrieved URLs were temporarily stored in a buffer of size 1500 to efficiently manage incoming data before processing. This buffered data was then utilized for feature extraction and incremental model updates. For the online learning phase, incoming URLs were processed in mini-batches of size $B = 500$. Incremental model updates were performed after each batch, enabling the framework to adapt to evolving phishing tactics in real time. Drift detection mechanisms were employed to monitor changes in feature distributions, and if drift was detected, a re-selection of features was initiated to maintain model performance. The Kolmogorov-Smirnov test, with a significance threshold of $\delta = 0.05$, was employed to detect feature drift by analyzing changes in the statistical distribution of features. Upon detecting significant drift, the system re-evaluated the feature set to ensure that the most relevant features were continuously selected.

5 | Results and Discussion

This section presents the experimental findings of our two-phase phishing URL detection framework, which combines offline training with online incremental learning. We first evaluate the effectiveness of our feature selection approach, demonstrating how LASSO regularization coupled with Mutual Information ranking identifies the most discriminative URL characteristics. We then analyzed the co-occurrence patterns of selected features through network visualization, revealing distinctive topological differences between malicious and benign URLs. The performance of various classification algorithms during the offline training phase is compared, followed by an examination of the online learning phase with a particular focus on adaptive feature selection and drift detection. Finally, we benchmark our approach against state-of-the-art phishing detection methods to demonstrate its superior accuracy, false positive rate, and adaptability to evolving threats.

To enhance the effectiveness of the initial feature set, a static feature selection strategy was employed, combining LASSO regularization with Mutual Information (MI) ranking. This approach yielded a subset of highly discriminative features capable of effectively distinguishing between phishing and legitimate URLs. Figure 5 illustrates the co-occurrence networks for malicious and benign URLs, visually representing the interrelationships among the selected features. To evaluate the classification performance using this optimized feature set, we implemented three distinct linear model variants-Linear Regression, Support Vector Machines (SVM), and the Perceptron-leveraging the flexibility of the SGD framework, which natively supports these algorithms.

The initial offline performance of three learning models-SGD-Perceptron, SGD-Logistic Regression (SGD-LR), and SGD-Support Vector Machine (SGD-SVM)-trained using features selected by a hybrid Lasso and Mutual Information (MI) approach is summarized in Tables 5–7. The experiments were conducted separately on the ISCX, Kaggle and PhiUSILL data sets to assess generalizability and robustness. Across all datasets, the SGD-SVM model consistently achieved the highest accuracy, precision, recall, and F1 score, indicating its superior ability to distinguish between phishing and legitimate URLs. These results also suggest that the Lasso and MI selected features effectively achieve high classification performance across the three data sets.

The performance of the SGD-based classifiers was evaluated on the combined data set, comprising URLs collected from ISCX, Kaggle, and PhiUSIIL, with the results summarized in Figure 6. Among the models, the SGD classifier configured with hinge loss (equivalent to a linear SVM) delivered the highest overall performance, achieving an accuracy of 99.85% and an exceptionally low False Positive Rate (FPR) of 0.067%. The Perceptron algorithm yielded comparable results with an accuracy of 99.73%, while the Logistic Regression implementation achieved 99.62% accuracy, demonstrating the efficacy of the selected features across multiple linear classifiers.

The line graph in Figure 7 illustrates the Kolmogorov-Smirnov statistic values for various features used in the Adaptive Feature Selection Pipeline for real-time phishing URL detection. This pipeline dynamically selects and updates features based on their importance and distribution changes, ensuring resilience against evolving phishing threats. The graph shows that features such as `url_length`, `url_path_length`, `token_count`, `num_params`, `num_dots`, `num_amps`, and `num_dots_path` exhibit high K–S statistic values (close to 1) with p–values of 0.0000, indicating significant distribution changes and a high

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TABLE 5 | Initial training performance with Lasso and MI-selected features on ISCX dataset.

SGD varient	Accuracy	Precision	Recall	F1-score	Training time (s)
SGD SVM	0.990150	0.983947	0.996278	0.990073	0.17382
SGD LR	0.995678	0.990412	0.998230	0.994304	0.19999
SGD Perceptron	0.990150	0.984102	0.996042	0.989020	0.17285

Note: All metrics are reported as decimal values. Training time is reported in seconds.

TABLE 6 | Initial training performance with Lasso and MI-selected features on Kaggle data set.

Model	Accuracy	Precision	Recall	F1-score	Training time (s)
SGD-Perceptron	0.989012	0.979261	0.957318	0.968641	0.1233
SGD-LR	0.979810	0.975672	0.936521	0.955684	0.1402
SGD-SVM	0.990000	0.989234	0.987674	0.988454	0.0918

Note: All metrics are reported as decimal values. Training time is reported in seconds.

TABLE 7 | Initial training performance with Lasso and MI-selected features on PhiUSIIL data set.

Model	Accuracy	Precision	Recall	F1-score	Training time (s)
SGD-Perceptron	0.9781	0.9572	0.9895	0.9730	1.5423
SGD-LR	0.9699	0.9483	0.9715	0.9597	1.5664
SGD-SVM	0.9878	0.9784	0.9873	0.9828	1.4439

Note: All metrics are reported as decimal values. Training time is reported in seconds.

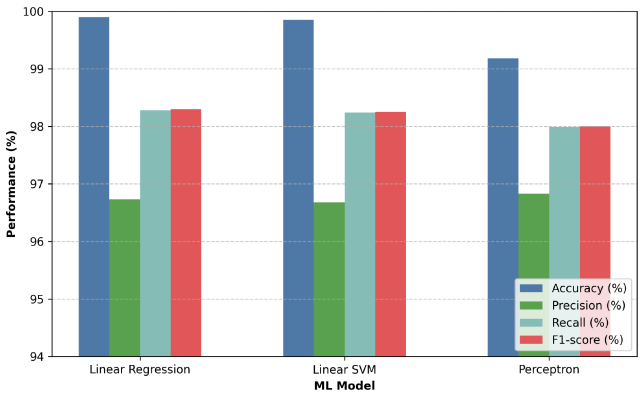


FIGURE 6 | Performance assessment of linear models on initial training dataset.

likelihood of being influenced by phishing attacks. These features are critical in distinguishing between phishing and legitimate URLs, prompting the algorithm to prioritize them during the adaptive selection.

Conversely, features like 'suspicious_file_extensions', 'num_percents', 'num_hyphen_host', 'num_delimiters_path', 'is_ip', 'url_shortening', and 'has_billing' show low K-S Statistic values with *p*-values above 0.05, suggesting that their distributions have not significantly changed. These features may contribute less to phishing detection or remain stable across different URL batches, leading the algorithm to deprioritize or retain them based on the drift detection results. The sharp peaks for certain features highlight moments of detected drift events, aligning with the Incremental Model Updates phase, where the Kolmogorov-Smirnov test is employed to monitor and respond to distribution shifts. The adaptive threshold mechanism, triggered by these drift detections, ensures that only the most relevant and evolving features are retained, enhancing the model's capability to detect sophisticated phishing tactics in real time. Following each major drift event, model performance initially experienced minor fluctuations but quickly stabilized or improved after feature re-selection.

Figure 8 shows the model performance metrics at different points during the online learning phase, demonstrating how adaptive feature selection maintains classification performance despite evolving phishing tactics.

Figure 9 provides a comparative overview of the performance metrics for Linear Regression, Linear SVM, and Perceptron models in both initial (offline) and updated (online) phases. Among the models, Linear SVM consistently achieved the highest performance across all metrics, demonstrating strong robustness and adaptability to online updates. Linear Regression maintained stable results with minimal performance degradation. In contrast, the Perceptron model exhibited a notable decline in accuracy, recall, and F1-score during the updated phase, though it showed improved precision. These results highlight Linear SVM as the most reliable choice for dynamic, real-time phishing detection scenarios.

To ensure resilience against evolving phishing tactics, our framework integrates an adaptive feature selection mechanism that updates the feature set based on distributional shifts in streaming data. Initially, the model utilized a static set of lexical and structural features selected through LASSO and Mutual

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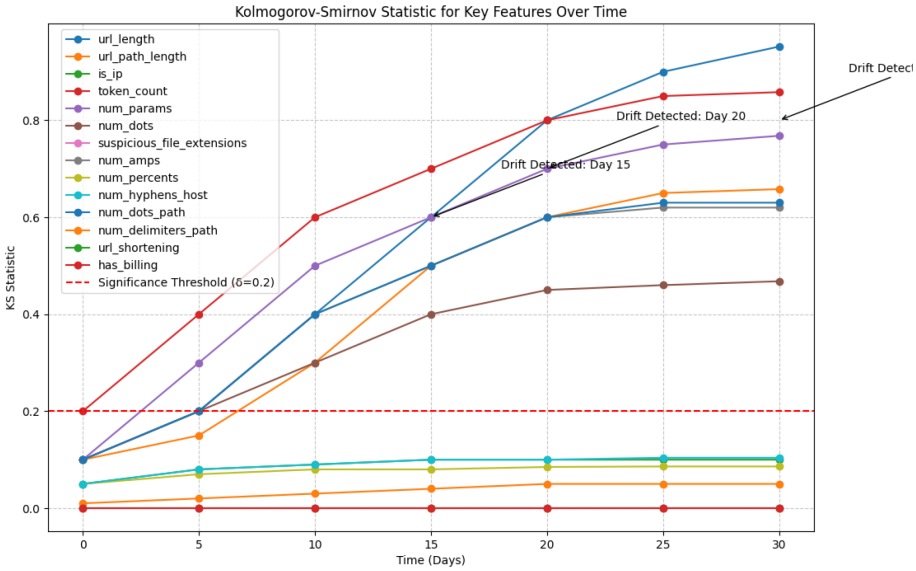


FIGURE 7 | Temporal analysis of feature distribution.

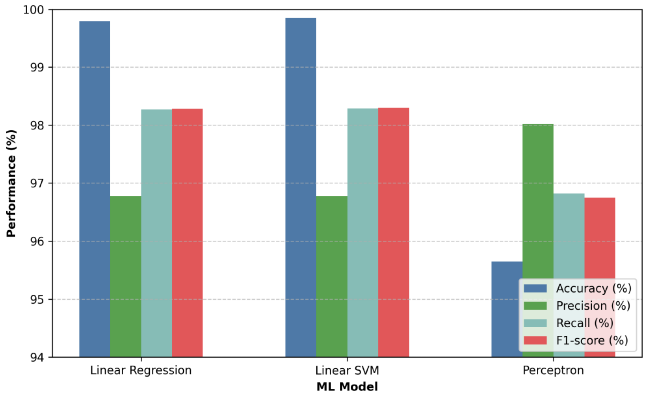


FIGURE 8 | Performance assessment of linear models after update.

Information. As new phishing patterns emerged, particularly in query obfuscation, the feature set was updated to include encoding- and query-related features. In the final phase, further shifts prompted the inclusion of behavioral and structural indicators such as 'suspicious_file_extensions' 'url_path_length' and 'num_ats'. This dynamic adaptation enables the model to sustain high detection accuracy in real-time environments. The selected features at each stage are summarized in Table 8.

5.1 | Ablation Study: Impact of Adaptive Feature Selection

To quantify the specific contribution of our adaptive feature selection mechanism, we conducted an ablation study comparing model performance with and without this component. We trained two variants of our best-performing model (Linear SVM):

1. Adaptive Model: Implementing full adaptive feature selection
2. Static Model: Using the same initial feature set throughout without adaptation.

Table 9 presents the results of our ablation study comparing performance with and without adaptive feature selection across different training data volumes. The adaptive model demonstrated consistently superior performance metrics, particularly at lower training volumes. With only 100 URLs for retraining, the adaptive model achieved an F1-score of 96.22%, compared to 92.31% for the static model. This performance gap remained significant with 500 URLs (97.15% vs. 93.89%), and remained present with 1000 URLs (98.30% vs. 93.73%).

5.2 | Runtime Performance Analysis

To comprehensively evaluate our framework's efficiency and demonstrate its advantages over traditional batch processing methods, we conducted extensive runtime performance testing. Specifically, we measured URL processing latency across different data sets and compared our incremental learning approach against conventional batch retraining methods.

5.2.1 | Per-URL Latency Analysis

Table 10 presents the average processing latency per URL for each data set using our SGD-SVM model, which consistently demonstrated superior classification performance. These results demonstrate that our framework can classify individual URLs with minimal latency, with average processing times between 12 to 15 ms across all datasets. This rapid classification capability is crucial for real-time phishing detection systems where immediate results are required.

5.2.2 | Comparison with Batch Processing Methods

To highlight the efficiency gains of our incremental learning approach, we compared it against traditional batch retraining methods that require complete model retraining when new data becomes available. As illustrated in Table 11, our incremental

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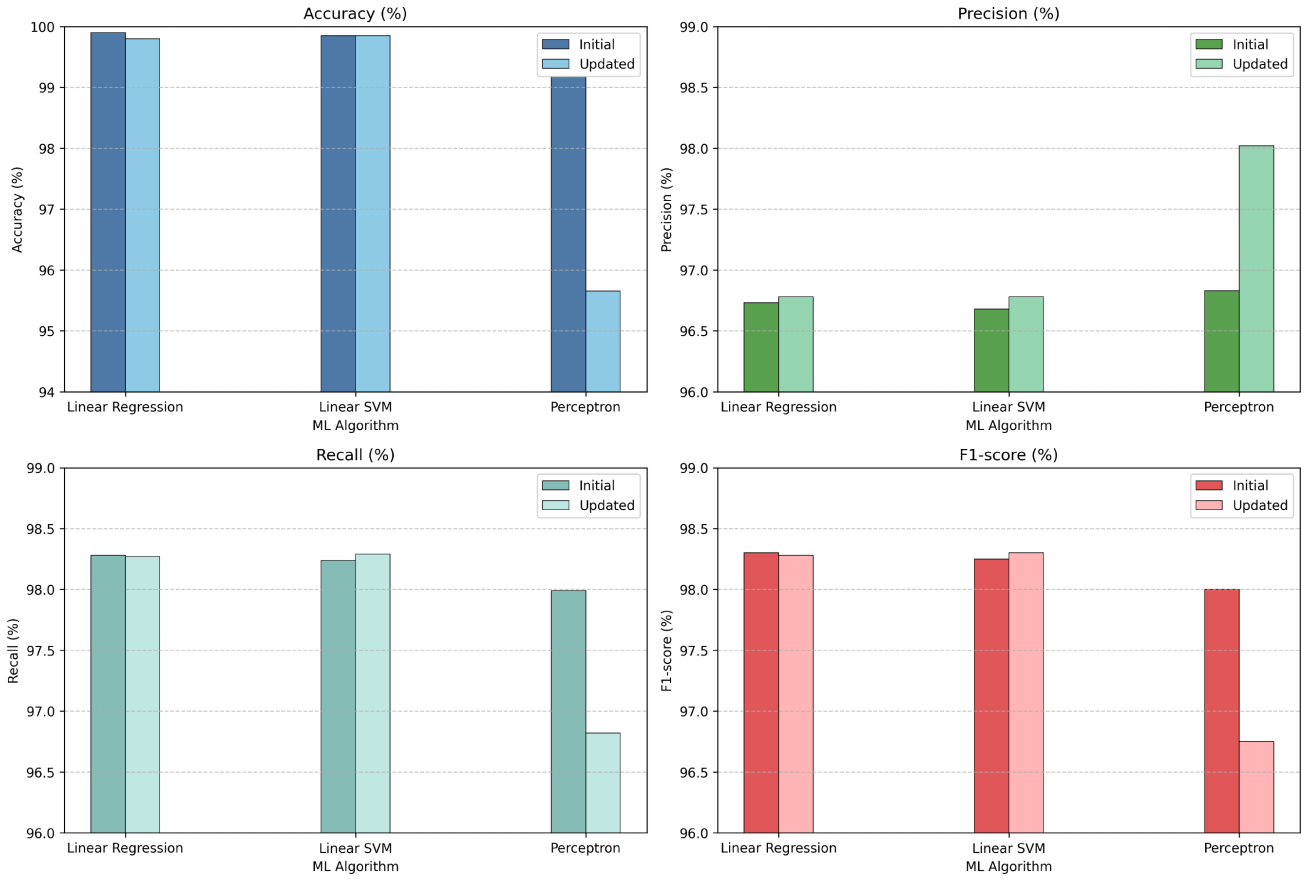


FIGURE 9 | Performance metrics comparison.

TABLE 8 | Evolution of adaptive feature sets for Phishing URL detection.

Time instance	Selected feature set
Initial stage	`url_length, url_path_length, is_ip, token_count, num_params, num_dots, suspicious_file_extensions, num_amps, num_percents, num_hyphens_host, num_digits_path, num_delimiters_path, has_billing`
Drift stage 1	`url_length, url_path_length, domain_length, token_count, num_digits, num_ats, num_subdirectories, suspicious_file_extensions, num_digits_query, non_alpha_numeric_chars, url_shortening, multiple_redirects, url_encoded, has_login`
Drift stage 2	`url_path_length, domain_length, token_count, num_digits, num_params, suspicious_file_extensions, homoglyph_detection, javascript_obfuscation, iframe_usage, redirects_chain_length, num_dots, num_ats, num_hyphens_host, url_shortening`

TABLE 9 | Ablation study results showing the impact of adaptive feature selection on linear SVM performance across varying training set sizes.

Model type	Training URLs	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Without adaptive features	100	99.15	91.12	93.54	92.31
With adaptive features	100	99.57	95.32	97.14	96.22
Without adaptive features	500	99.15	92.78	95.02	93.89
With adaptive features	500	99.79	96.45	97.86	97.15
Without adaptive features	1000	99.15	92.65	94.83	93.73
With adaptive features	1000	99.85	96.78	98.29	98.30

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TABLE 10 | Per-URL processing latency metrics.

Dataset	Avg. latency (ms)	Median latency (ms)	95th percentile (ms)	URLs processed
ISCX	12.4	11.8	18.3	11 430
Kaggle	13.7	12.9	21.5	10 000
PhiUSIIL	15.2	14.6	25.7	34 560
Combined	14.3	13.5	22.8	55 990

Note: All latency measurements reported in milliseconds (ms).

TABLE 11 | Runtime comparison between incremental and batch processing.

Operation	Incremental approach (ms)	Batch retraining (ms)	Efficiency gain (×)
Model update (100 URLs)	78.5	1247.6	15.9×
Model update (500 URLs)	196.3	5832.4	29.7×
Model update (1000 URLs)	342.8	11 568.9	33.7×
Feature adaptation	124.7	3856.2	30.9×

Note: Efficiency gain is calculated as the ratio of batch processing time to incremental processing time. All operations were repeated 10 times and averaged to ensure statistical significance.

learning approach demonstrates remarkable efficiency advantages over batch methods, with performance gains of 15–33× depending on the batch size. The efficiency gap widens significantly as the number of URLs increases, highlighting the scalability of our approach.

5.3 | Model Adaptation Dynamics: Effectiveness and Limitations in Responding to Concept Drift

While the adaptive feature selection strategy demonstrated effectiveness in capturing evolving patterns, its responsiveness to abrupt concept drift remains limited, potentially impacting detection accuracy in rapidly changing threat landscapes. As observed in Figure 6, features like `url_length`, `url_path_length`, and `token_count` exhibited high K–S statistic values, indicating significant distribution changes that our model successfully adapted to. However, the adaptation process required multiple data batches before stabilizing, suggesting a potential vulnerability window during sudden tactical shifts by attackers.

This limitation was particularly evident following major drift events, where although model performance ultimately stabilized as seen in Figure 9, the initial response period showed performance fluctuations. The Linear SVM model, while performing consistently better than Linear Regression and Perceptron models, still demonstrated a brief lag in adaptation when facing abrupt changes in feature distributions rather than gradual shifts. The feature set updates summarized in Table 8 reflect our model's eventual adaptation to new phishing patterns, but the temporal analysis suggests that sophisticated attackers could potentially exploit the adaptation window by rapidly switching tactics. This highlights the need for complementary detection mechanisms specifically designed to identify and respond to abrupt concept drift in real-time threat environments.

5.4 | Labeling Constraints in Operational Environment

Our experimental evaluation assumes the availability of labeled streaming data for adaptation, which represents an ideal scenario that may not fully reflect operational realities. In practice, obtaining timely labels for phishing URLs often involves manual verification processes, creating inevitable delays between URL observation and confirmed labeling. These delays could potentially impact the system's ability to adapt quickly to emerging phishing tactics. To address this practical constraint, several semi-supervised and active learning approaches could enhance our framework's applicability. A confidence-based self-training mechanism could leverage high-confidence predictions as pseudo-labels during periods of labeling unavailability, maintaining adaptation capabilities while awaiting expert verification. Additionally, uncertainty-based active learning strategies could optimize limited labeling resources by prioritizing the most informative samples-particularly those near decision boundaries or exhibiting characteristics of potential drift-for expert review. Such approaches would create a more robust adaptation pipeline that balances the need for labeled data with operational realities. Incorporating these techniques represents a natural extension of our adaptive feature selection framework that would further enhance its practical deployment in real-world security environments where continuous access to fully labeled data cannot be guaranteed.

5.5 | Comparison of Proposed Method with the State of the Art

URL features with powerful text embedding transformers and DL model features.

A comparison of the proposed method with the existing literature study is given in Table 12. The proposed work demonstrates several advantages over existing state-of-the-art phishing detection methods by incorporating third-party independence (TPI), adaptive feature selection, and an online learning approach. Unlike many batch learning approaches that rely on third-party services for URL classification, such as Sahingoz et al. [15] and Jain & Gupta et al. [17], the proposed method is fully independent,

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TABLE 12 | Comparative analysis of state-of-the-art phishing detection approaches.

References	Feature categories	Feature count	TPI	Adaptive feature selection	Learning paradigm	FPR (%)	Accuracy (%)
Sahingoz et al. [15]	Lexical (NLP-based)	—	✓	×	Batch	2.95	97.98
Jain & Gupta [17]	URL, Hyperlink	19	✓	×	Batch	1.25	99.09
Jain & Gupta [19]	Hyperlink	12	×	✓	Batch	1.52	98.40
Das Gupta et al. [20]	URL, Hyperlink	24	✓	—	Batch	0.0049	99.17
Moghimi et al. [26]	URL, Web Content	—	✓	×	Batch	—	99.14
Alsarier et al. [27]	HTML, JavaScript, Domain, Address Bar	—	×	×	Batch	0.28	97.00
Zamir et al. [28]	URL, Hyperlink, Web Content	—	×	×	Batch	—	97.40
Azeez et al. [29]	URL, Web Content	—	✓	×	Batch	—	99.10
Jalal et al. [30]	URL	—	✓	×	Batch	3.70	96.25
Satheeshkumar et al. [31]	URL, Domain, Web Content	—	×	×	Batch	1.50	97.13
Aljofey et al. [1]	URL, HTML	—	✓	×	Batch	2.09	98.48
Prasad A et al. [21]	URL, HTML, Derived	—	✓	×	Incremental	—	99.79
Sudar et al. [32]	URL, Human Engineered, Web based	27	×	×	Batch	—	98.66
Sudar et al. [33]	URL (handcrafted), Text Embedded, DL Derived	—	✓	×	Batch	—	98.31
Proposed Method	**Lexical**	**40**	✓	✓	**Online**	**0.067**	**99.85**

Note: TPI: Third-Party Independence; FPR: False Positive Rate; —: Information not specified in the original publication. Feature Count refers to the total number of features utilized in the respective approach (when specified).

allowing for faster and more reliable real-time detection without external dependencies. Additionally, the false positive rate (FPR) of the proposed work, at 0.067%, is significantly lower than methods like Sahingoz et al. [15] (2.95%), and Aljofey et al. [1] (2.09%), reducing the number of benign URLs misclassified as phishing. This low FPR is crucial for improving user trust and system efficiency by minimizing unnecessary alarms.

While a few prior works such as Prasad et al. [21] report better accuracy (99.79%), it is essential to contextualize these results in light of additional critical performance metrics such as false positive rate (FPR), learning paradigm, and feature efficiency. Our method achieves a significantly lower FPR (0.067%), which is crucial for minimizing disruption in real-world applications. Furthermore, unlike Prasad et al., whose method relies on webpage-derived features requiring the entire webpage to be loaded and parsed, our approach is based solely on lexical features that are immediately available from the URL. This results in a substantial improvement in real-time detection capability and response time, making our model better suited for scalable, real-time security systems. Additionally, our model supports online learning, enabling continuous updates without the need for full retraining, thereby offering superior adaptability to emerging threats.

The comparative results illustrated in Figure 10 demonstrate the superiority of the proposed method over existing studies. Specifically, the proposed approach achieves the highest classification accuracy while maintaining the lowest false positive rate (FPR), highlighting its effectiveness in correctly identifying phishing URLs with minimal misclassification. Moreover, the adaptive feature selection in the proposed method offers a dynamic response to evolving phishing attacks, unlike the static approaches seen in most batch-learning methods, allowing for better generalization and real-time adaptability. These innovations and the online learning approach make the proposed system more responsive and accurate for phishing detection in real-time environments.

5.6 | Limitations of the Study

While our approach demonstrates significant advancements in phishing detection, certain constraints merit acknowledgement. Unlike content-based and HTML-based phishing detection approaches, the URL-based detection methodology does not analyze the actual webpage content, scripts, or visual elements, which may limit its effectiveness against highly obfuscated or visually deceptive phishing attacks that rely on content-based deception. Even though the content-based and HTML-based phishing detection approaches can capture detailed behavioral and structural characteristics of malicious webpages, they demand substantial computational and memory resources. These methods typically require full webpage rendering, DOM tree analysis, and sometimes JavaScript execution, making them unsuitable for real-time detection in large-scale or resource-constrained environments.

While our approach provides substantial advantages in terms of efficiency, scalability, and independence from third-party services, it may fall short in handling some obfuscation strategies. For instance, attackers employing highly obfuscated scripts, particularly those using homoglyphs in non-Latin writing systems, pose detection challenges. To address this, we integrated enhanced homoglyphs detection, yet limitations may persist in incredibly nuanced or novel encoding schemes. Another constraint is that URL-based methods alone may not reliably detect phishing attacks that exploit legitimate domains but embed malicious content. To mitigate this, our model includes specialized lexical features such as data URIs and JavaScript to identify such embedded threats.

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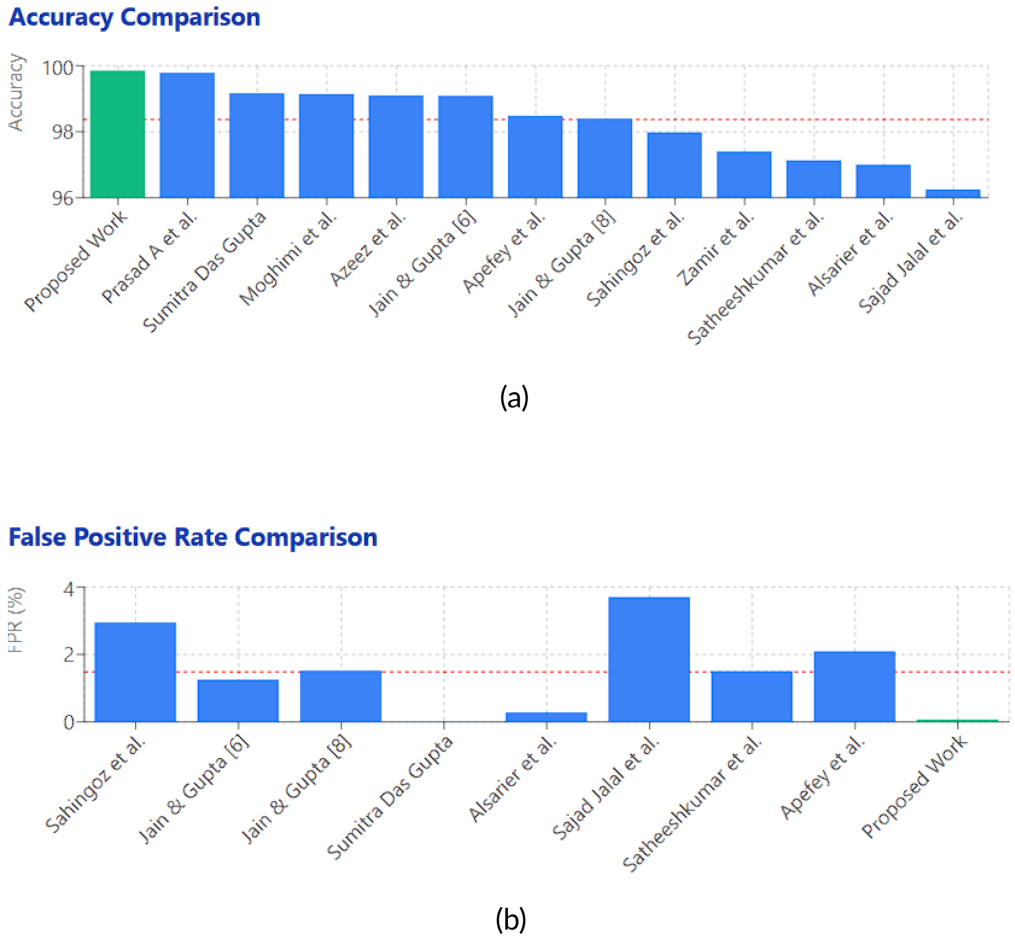


FIGURE 10 | Comparative evaluation of the proposed method and existing studies. (a) Accuracy, (b) False positives.

Also, context-dependent phishing attacks, such as spear-phishing, often involve highly personalized content and may not exhibit obvious lexical anomalies in the URL alone. As such, a lexical-only framework might fail to detect these nuanced threats. While our current approach is designed for lightweight, real-time detection based on URL characteristics-prioritizing speed, scalability, and third-party independence-it does trade off some contextual understanding that hybrid methods could offer.

While hybrid approaches that integrate lexical, contextual, and behavioral feature dimensions can overcome single-vector evasion techniques, they simultaneously introduce increased computational complexity that significantly impacts detection latency-critical factor in effective phishing prevention where milliseconds can determine successful user protection. This latency constraint is particularly crucial at the enterprise scale, where high-volume traffic requires near-instantaneous classification decisions. Moreover, hybrid models face potential feature interdependence challenges where correlations between different feature types may create redundancies or conflicting signal patterns that can degrade overall model performance if not appropriately addressed.

In contrast, our proposed lexical feature-based framework avoids the overhead associated with content rendering by focusing solely on URL characteristics. This design choice enables faster inference, lower memory usage, and seamless integration at early detection stages, such as browser extensions. However, this efficiency comes with trade-offs: Our model may miss attacks that rely purely on content-based deception, such as visual impersonation or malicious script execution, that is not reflected in the URL. Future work may explore hybrid models that incorporate lightweight content inspection selectively, applied only to high-risk or ambiguous URLs, thereby preserving scalability while improving detection coverage.

6 | Conclusion and Future Work

In this study, we proposed an efficient and adaptive phishing URL detection framework that integrates online learning with dynamic feature selection techniques. By leveraging third-party-independent lexical features, the framework ensures lightweight and scalable detection without reliance on external data sources. Our approach combines an initial offline phase—LASSO regularization and Mutual Information ranking for optimal feature selection—with a robust online learning phase guided by Kolmogorov-Smirnov drift detection. This dual-phase strategy enables the model to adapt seamlessly to evolving phishing patterns by incorporating incremental updates through mini-batch gradient learning, thereby avoiding costly retraining. Temporal analysis of the PhishTank data set (June

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2024–January 2025) revealed significant distributional shifts in critical URL features, substantiating the necessity for adaptive defenses. The framework's ability to process streaming data in real time, coupled with its high detection accuracy (99.85%) and exceptionally low false positive rate (0.067%), underscores its effectiveness in countering emerging phishing attacks. These results validate both the discriminative power of the selected features and the resilience of the proposed model, establishing it as a practical and high-performance solution for dynamic phishing threat environments.

Although our adaptive feature selection approach demonstrates promising results for real-time phishing URL detection, several important research directions remain for future exploration. Future research should explore hybrid approaches that combine our adaptive feature selection with ensemble methods or drift detection mechanisms specifically tuned for abrupt changes. This could include incorporating a secondary rapid-response model that activates during detected periods of high volatility or implementing a more aggressive feature re-weighting scheme during periods of suspected tactical shifts by attackers. Additionally, exploring the use of anomaly detection as an early warning system for abrupt concept drift could reduce the vulnerability window we identified in our current approach.

For resource-constrained environments such as IoT devices, optimized implementations with streamlined feature extraction and distributed detection architectures would extend our approach's practical applicability. Furthermore, systematic evaluation against adversarial manipulation through sophisticated evasion techniques would strengthen the system's resilience, particularly against gradual feature manipulation designed to bypass drift detection. Finally, assessing performance against AI-generated and zero-day phishing URLs will validate effectiveness in emerging threat landscapes, especially against attacks designed to exploit the limitations we've identified in our current approach. This includes exploring feature extraction optimizations, model compression techniques, and distributed detection architectures where feature selection could be performed on edge gateways while maintaining detection capabilities on constrained devices. Additionally, evaluating power consumption implications of continuous adaptation in battery-operated devices represents an important practical consideration.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data sets used in the paper are publicly available.

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